Theoretical Computer Science

Lecture Notes

Prof. Dr. Titus Dose

OTH Regensburg – Faculty of Computer Science and Mathematics

October 16, 2024

Table of Contents I

1. Mathematical Basics
1.1 Foundations from Propositional Logic
1.2 Methods of Proof
1.3 Basic Notations and Terms

Table of Contents II

2. Computability
2.1 History of the Concept of Algorithms
2.2 Alphabets, Words, Formal Languages
2.3 While Programs
2.4 Register Machines
2.5 Mini-While Programs

Table of Contents III

1
Mathematical Basics

1.1
Foundations from Propositional Logic

This lecture is about proving mathematical statements.
A mathematical statement is either true (1) or false (0). So it always has a unique truth value.

Some examples:

- 5 is a prime
- there are infinitely many twin primes (a twin prime is a pair $(p, p+2)$ such that both numbers are prime)
- there are no even primes

Can the following sentence be considered as a statement?
"This is a sentence, which is wrong"

We will be using the following logical connections that turn statements into more complicated statements:

- "and", conjunction, $\wedge$
- "or", disjunction, $\vee$
- "not", negation, $\neg$
- implication, $\Rightarrow$, "if ..., then ..."
- equivalence, $\Leftrightarrow$, "if and only if ..., then ..."

In general, the natural language description should be preferred to the formula notation using symbols: e.g., "and" should be preferred to "$\wedge$".

The following truth value table *defines* the connections:

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \Rightarrow B$	$A \Leftrightarrow B$
1	1	0	1	1	1	1
1	0	0	0	1	0	0
0	1	1	0	1	1	0
0	0	1	0	0	1	1

Some typical mistakes and misunderstandings:

- the negation of the statement "Each prime is odd" is **not** "Each prime is even", but "There is an even prime".
- the disjunction is inclusive (contrary to "either-or"), i.e., the following statement is true: "2 is even or 2 is a prime".
- Everything is implied by a false statement. E.g., the following statement is true: "If 9 is a prime, then there are infinitely many twin primes" (even if we do not know whether there are infinitely many twin primes).[1]

[1] This is in accordance with everyday language: the statement "If it rains tomorrow, then the street will get wet" is considered to be true even if it does not rain the next day.

- Note: $A \Leftrightarrow B$ is equivalent with $(A \Rightarrow B) \wedge (B \Rightarrow A)$

A	B	$A \Rightarrow B$	$B \Rightarrow A$	$(A \Rightarrow B) \wedge (B \Rightarrow A)$	$A \Leftrightarrow B$
1	1	1	1	1	1
1	0	0	1	0	0
0	1	1	0	0	0
0	0	1	1	1	1

A typical mistake is to forget one of the two implications, especially when having a longer chain of equivalent statements. As an example, consider the following statements about a real number x

$$\begin{aligned} x = 2 &\Leftrightarrow x + 1 = 3 \\ &\Leftrightarrow (x+1)^2 = 9 \text{ WRONG!} \\ &\Leftrightarrow 2^{(x+1)^2} = 512 \end{aligned}$$

However, the statements "$x = 2$" and "$2^{(x+1)^2} = 512$" are not equivalent as $-4 \neq 2$, but $2^{(-4+1)^2} = 512$.

Tip: In the first semester, mainly do without $\Leftrightarrow$ and instead prove $A \Rightarrow B$ and $B \Rightarrow A$ separately.

1.2
Methods of Proof

We will be using 3.5 methods of proof.

1. Direct proof
2. Indirect proof
   - 2.1 Proof by contradiction
   - 2.2 Proof by contraposition
3. Induction

Direct Proof

Whenever we prove a mathematical statement, we start with a statement A, assume it to be true, and show that then another statement B is also true (i.e., we always prove implications).

The condition A is not always stated explicitly. We can e.g. prove the statement "2 is a prime". But where's the premise?

When proving the statement, we will have to use some premises, e.g., certain properties of the multiplication of natural numbers, properties of natural numbers,...
We cannot squeeze water from a stone. We always need axioms.

Direct proof: we assume that A is true and show by a chain of logical inferences[2] that B is true.

[2] It can be proven that a couple of simple inference rules suffice. Thus, it is easy to write a program that can check arbitrary mathematical proofs for their correctness (if they are written down detailed enough).

Direct Proof – Example

Let p be a natural number. Let A be the statement "p is an even prime" and B the statement "p is less than 5". We now prove $A \Rightarrow B$.

Proof.

We assume that A is true. So p is an even prime.
A implies statement C: "$p = 2$" because every other even number is divisible by 2 and thus no prime.
C implies B as 2 is less than 5. $\square$

Contraposition

The statement $A \Rightarrow B$ is equivalent to $\neg B \Rightarrow \neg A$.[3]

Instead of proving $A \Rightarrow B$ directly, we give a direct proof for the statement $\neg B \Rightarrow \neg A$.

Example: Let p be a natural number. Let A be the statement "p^2 is not divisible by 4" and B the statement "p is odd". We prove $A \Rightarrow B$.

Proof.

We give a proof by contraposition.
So we start with the statement $\neg B$: "p is even".
Then $p = 2 \cdot q$ for some natural number q.
Then $p^2 = (2 \cdot q)^2 = 4 \cdot q^2$, so p^2 is divisible by 4, which is the statement $\neg A$. $\square$

[3] Write down the truth value table if you are not convinced. Cf. page 10.

Proof by Contradiction

The proof by contradiction is similar to the proof by contraposition.

We prove $A \Rightarrow B$ by starting with the statement $A \wedge \neg B$ and inferring a contradiction. Then our assumption $A \wedge \neg B$ must have been wrong (otherwise, it wouldn't imply a contradiction).

Thus, it is impossible that A is true and B is not. In other words, if A is true, then B is also true, i.e., $A \Rightarrow B$.

Proof by Contradiction – Example

Example: Let us prove that every natural number greater than 1 has a divisor that is prime (prime divisor).

Proof.

Let's assume for a contradiction that there exists some number that does not have a prime divisor. Let p be the least such number.

Then p is no prime (p is divisor of p). Thus, $p = r \cdot s$ for natural numbers $r, s > 1$.

As p is the least number > 1 without any prime divisors, $r > 1$, and $r < p$, the number r has a prime divisor t, i.e., $r = t \cdot u$ for a natural number u.

But then $p = r \cdot s = t \cdot u \cdot s$ and since t is prime, p has a prime divisor. This is a contradiction. So our assumption was wrong and every natural number > 1 has a prime divisor. $\square$

Proof by Contradiction – 2nd Example

Now we can prove with the same technique that there are infinitely many primes.

Proof.

Let's assume for a contradiction that there exist only finitely many primes, i.e., there are some $n \in \mathbb{N}$ and primes $p_1, \ldots, p_n$ such that these are the only primes.

Consider $p = p_1 \cdot p_2 \cdot \cdots \cdot p_n + 1$.
As p is greater than 1, it has a prime divisor (by the statement proven on the last page). So there is some i so that p_i is a divisor of p, i.e., $p = p_i \cdot s$ for a natural number s.
Thus $s = \frac{p}{p_i} = \frac{p_1 \cdot p_2 \cdot \cdots \cdot p_n + 1}{p_i} = \frac{p_1 \cdot p_2 \cdot \cdots \cdot p_n}{p_i} + \frac{1}{p_i}$, i.e., $\frac{1}{p_i} = s - \frac{p_1 \cdot p_2 \cdot \cdots \cdot p_n}{p_i}$.
So $\frac{1}{p_i}$ is the difference of two natural numbers and hence, an integer itself. However, as p_i is prime, $0 < \frac{1}{p_i} \leq \frac{1}{2}$, a contradiction. $\square$

Complete Induction

Complete Induction is a method of proof we will be using extensively.

Consider the statement: For all natural numbers n the following statement A_n is true:

$$\sum_{i=1}^{n} i = \frac{n^2 + n}{2} \quad .^4$$

Proving all statements A_n for $n \in \mathbb{N}$ directly is not straightforward. The idea of complete induction is to prove the statement A_0 (base case, BC) and "for all $k \in \mathbb{N}$, if A_k is true, then so is A_{k+1}" (induction step, IS). This proves that all A_n are true:

$$A_0 \Rightarrow A_1 \Rightarrow A_2 \Rightarrow A_3 \Rightarrow A_4 \Rightarrow \dots$$

Note: BC and IS can be proven using any of the proof methods.

[4] $\sum_{i=1}^{n} i = 1 + 2 + \cdots + (n-1) + n$

Complete Induction – Example

For all natural numbers n the following statement A_n is true:

$$\sum_{i=1}^{n} i = \frac{n^2 + n}{2}$$

Proof.

BC: We prove A_0: It holds $\sum_{i=1}^{0} i = 0 = \frac{0^2+0}{2}$.

IS: We prove "for all $k \in \mathbb{N}$, if A_k is true, then so is A_{k+1}". Let $k \in \mathbb{N}$. We assume that A_k is true (**induction hypothesis**). So $\sum_{i=1}^{k} i = \frac{k^2+k}{2}$.
In order to complete the proof, we need to prove A_{k+1}

Recall: We have assumed the statement A_k:

$$\sum_{i=1}^{k} i = \frac{k^2+k}{2}.$$

We want to prove the statement A_{k+1} (in order to complete IS):

$$\sum_{i=1}^{k+1} i = \frac{(k+1)^2+k+1}{2}.$$

Then

$$\sum_{i=1}^{k+1} i = k+1+\underbrace{\sum_{i=1}^{k} i}_{\overset{A_k}{=}\frac{k^2+k}{2}} = k+1+\frac{k^2+k}{2} = \frac{2(k+1)+k^2+k}{2}$$

$$= \frac{k^2+2k+1+k+1}{2} = \frac{(k+1)^2+k+1}{2}$$

holds[5]. Thus A_{k+1} holds, IS is proven, and the proof is complete. $\square$

[5]Note: the last equation is due to Francesco Binomi (1727 – 1643).

Inductive Definitions

Having introduced the concept of proofs via complete induction, we can now use the same concept for definitions (inductive definition).

We define the summation operation $\sum$. Let $s_1, s_2, \ldots$ be natural numbers. We define:

BC: $\sum_{i=1}^{0} s_i := 0$.

IS: $\sum_{i=1}^{k+1} s_i := s_{k+1} + \sum_{i=1}^{k} s_i$.

Why does this define e.g. the expression $\sum_{i=1}^{5} i$?

$$\sum_{i=1}^{5} i \overset{\text{IS}}{=} 5 + \sum_{i=1}^{4} i \overset{\text{IS}}{=} 5 + 4 + \sum_{i=1}^{3} i \overset{\text{IS}}{=} 5 + 4 + 3 + \sum_{i=1}^{2} i$$

$$\overset{\text{IS}}{=} 5 + 4 + 3 + 2 + \sum_{i=1}^{1} i \overset{\text{IS}}{=} 5 + 4 + 3 + 2 + 1 + \sum_{i=1}^{0} i$$

$$\overset{\text{BC}}{=} 5 + 4 + 3 + 2 + 1 + 0.$$

Structural Induction

Inductive proofs are not only useful when considering statements that hold for all natural numbers.

Generally, they can be used for recursively/inductively defined structures. E.g., for **polynomials** over the integers with one variable:

BC Let x be a variable. Then

- x is a polynomial
- each integer z is a polynomial

IS If p_1 and p_2 are polynomials over the integers with one variable x, then $(p_1 + p_2)$ and $(p_1 \cdot p_2))$ are polynomials.

Example:[6] $((5 \cdot ((((x \cdot x) \cdot x) - (3 \cdot (x \cdot x))) + 12)) \cdot ((((3 \cdot x) - 5))))$
5, x, 3, 12 are polynomials by BC. Applying IS iteratively finally leads to the polynomial above.

[6] Note: later on we will omit unnecessary brackets and write $5 \cdot (x^3 - 3x^2 + 12) \cdot (3x - 5)$.

Structural Induction – Example

Let's prove: All polynomials have the same number of opening and closing brackets.

Proof.

BC: Neither x nor an integer contains any bracket.

IS: Let p_1 be polynomial with a opening and a closing brackets. Let p_2 be polynomial with b opening and b closing brackets.

The polynomial $(p_1 + p_2)$ has $a + b + 1$ opening and closing brackets. The same holds for $(p_1 \cdot p_2)$. This completes the proof.

In the following we use the term "induction" for both "complete induction" and "structural induction". $\square$

1.3
Basic Notations and Terms

We use ( ) and [ ] to bracket mathematical expressions. Arguments of functions are placed in ( ). Sets are bracketed with $\{\ \}$.

$$
\begin{aligned}
\mathbb{N} &\stackrel{df}{=} \{0, 1, \ldots\} = \text{Set of natural numbers} \\
\mathbb{N}^+ &\stackrel{df}{=} \{1, 2, \ldots\} = \text{Set of positive natural numbers} \\
\mathbb{P} &\stackrel{df}{=} \{2, 3, 5, 7, 11, 13, 17, \ldots\} = \text{Set of primes} \\
\mathbb{Z} &\stackrel{df}{=} \{\ldots, -2, -1, 0, 1, 2, \ldots\} = \text{Set of integers} \\
\emptyset &\stackrel{df}{=} \text{empty set}
\end{aligned}
$$

Important: **0 is a natural number.**

For $x \in \mathbb{N}$, $y \in \mathbb{N}^+$ let $(x \bmod y)$ denote the remainder of the division x/y, i.e., $(x \bmod y) = x - zy$, where $z \in \mathbb{N}$ is the greatest number with $zy \leq x$.

Sets

Notation for the definition of sets:

$$\begin{aligned}\{n : n \in \mathbb{N} \text{ and } n \geq 5\} &= \{n \in \mathbb{N} : n \geq 5\} \\ &= \{n \mid n \in \mathbb{N} \text{ and } n \geq 5\} \\ &= \{n \in \mathbb{N} \mid n \geq 5\} \\ &= \{5, 6, \ldots\}\end{aligned}$$

Element relationship and inclusion:

- $a \in M \overset{df}{\Longleftrightarrow}$ a is an element of the set M
- $a \notin M \overset{df}{\Longleftrightarrow}$ a is no element of the set M
- $M \subseteq N \overset{df}{\Longleftrightarrow}$ for all a, if $a \in M$, then $a \in N$ (M is subset of N)
- $M \nsubseteq N \overset{df}{\Longleftrightarrow}$ it does not hold $M \subseteq N$ (M is no subset of N)
- $M \subsetneq N \overset{df}{\Longleftrightarrow}$ $M \subseteq N$ and $M \neq N$ (M is proper subset of N)

Set Operations

$$
\begin{array}{rcll}
A \cap B & \stackrel{df}{=} & \{a \mid a \in A \text{ and } a \in B\} & \text{(Intersection of } A \text{ and } B\text{)} \\
A \cup B & \stackrel{df}{=} & \{a \mid a \in A \text{ or } a \in B\} & \text{(Union of } A \text{ and } B\text{)} \\
A \setminus B & \stackrel{df}{=} & \{a \mid a \in A \text{ and } a \notin B\} & \text{(Difference of } A \text{ and } B\text{)} \\
A - B & \stackrel{df}{=} & A \setminus B & \text{(Difference of } A \text{ and } B\text{)} \\
\overline{A} & \stackrel{df}{=} & M \setminus A & \text{(Complement of } A \text{ relative} \\
 & & & \quad \text{to a fixed base set } M\text{)} \\
\mathcal{P}(A) & \stackrel{df}{=} & \{B \mid B \subseteq A\} & \text{(Power set of } A\text{)}
\end{array}
$$

$\#A = |A| \stackrel{df}{=}$ number of elements of a finite set A

Tuple (Vector) and Cartesian Product

For $n \in \mathbb{N}$ we define the following.

- $(a_1, a_2, \ldots, a_n) \stackrel{df}{=}$ sequence of elements $a_1, a_2, \ldots, a_n$ in this order (n-tuple, n-dimensional vector)
- $A_1 \times A_2 \times \ldots \times A_n \stackrel{df}{=} \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for all } i\}$ (Cartesian product of sets A_1,$A_2, \ldots, A_n$)
- $A^n \stackrel{df}{=} \underbrace{A \times A \times \ldots \times A}_{n \text{ times}}$ (n-dim. Cartesian product of set A)

The first definition yields the empty tuple () for $n = 0$. Thus $A^0 = \{()\}$ and $|A^0| = 1$.

Quantifiers

Quantifiers are merely abbreviations:

- $\exists =$ "there exist(s)"
- $\forall =$ "for all"

Instead of "Every even natural number greater or equal 4 is the sum of two (non-necessarily distinct) primes" we may write as shortcut $\forall_{n \in \{2 \cdot i \mid i \in \mathbb{N}, i \geq 2\}} \exists_{p,q \in \mathbb{P}}\ n = p + q$.

For better readability, we will go without subscripts when using quantifiers, i.e., $\forall n \in \{2 \cdot i \mid i \in \mathbb{N}, i \geq 2\} \exists p, q \in \mathbb{P}\ n = p + q$.

Negating such statements can be done mechanically, replace $\forall$ with $\exists$ and vice versa, and negate the remaining statement: $\exists n \in \{2 \cdot i \mid i \in \mathbb{N}, i \geq 2\} \forall p, q \in \mathbb{P}\ n \neq p + q$.

Functions

A function f from A to B (notation $f : A \to B$) is determined by the source set (aka domain) A, the target set (aka codomain) B, and the graph $G_f \subseteq A \times B$, where for every $a \in A$ there is at most one $b \in B$ with $(a, b) \in G_f$.

If $(a, b) \in G_f$, then the function f at the position a has the function value b (notation: $f(a) = b$).

If there is no $b \in B$ with $f(a) = b$, then $f(a)$ is not defined (notation: $f(a) = \text{n.d.}$).

Example: The functions g and h are different, although $G_g = G_h$.

- $g : \mathbb{N} \to \mathbb{N}$ with $g(x) = 0$ for all $x \in \mathbb{N}$
- $h : \mathbb{Z} \to \mathbb{Z}$ with $h(x) = 0$ if $x \geq 0$ and $h(x) = \text{n.d.}$ otherwise

Definition 1.1

Let $f\colon A \to B$ and $g\colon B \to C$ be functions.

- $\boldsymbol{g \circ f}$ denotes the function $A \to C$ with $(g \circ f)(x) \stackrel{df}{=} g(f(x))$ (composition of functions).
- **Domain of definition** of f: $D_f \stackrel{df}{=} \{a \in A \mid \exists b \in B\ f(a) = b\}$
- **Range** of f: $R_f \stackrel{df}{=} \{b \in B \mid \text{there is } a \in A \text{ with } f(a) = b\}$
- f is **total** $\stackrel{df}{\iff} D_f = A$
- f is **surjective** $\stackrel{df}{\iff} R_f = B$
- f is **injective** $\stackrel{df}{\iff} f(a_1) \neq f(a_2)$ for all distinct $a_1, a_2 \in D_f$
- f is **bijective** $\stackrel{df}{\iff} f$ is total, surjective, and injective
- If f injective, there exists the **inverse function** $f^{-1}\colon B \to A$ with $\boldsymbol{f^{-1}(b)} \stackrel{df}{=}$ the $a \in A$ with $f(a) = b$.
  Note: for an injective function f it holds $f^{-1}(f(a)) = a$ for $a \in D_f$ and $f(f^{-1}(b)) = b$ for $b \in R_f$.

2
Computability

Computability – Outline

- Everyone has a certain idea of what computers can do. We want to take a closer look at this question and find out which tasks can in principle be solved with computers. This requires meaningful and precise definitions of the terms *algorithm* and *computability*. The creation of these definitions is considered one of the most important contributions to mathematics in the 20th century and was the result of a development that can be traced back to discussions of philosophical questions in antiquity.
- In this chapter, we will first get to know various computation models: While programs (WHILE), Random access machines (RAM), Turing machines (TM).

Computability – Outline II

- We show the equivalence of these computational models and thus substantiate the thesis that they capture exactly what computers can do.
- We have now defined the terms *algorithm* and *computability* in a meaningful and precise way.
- Advantage: We can now also recognize the limits of computability. We show that many easily formulated tasks cannot be solved by computers.
- For example, the question of whether two given computer programs compute the same function cannot be solved by computers. This shows that the automatic verification of arbitrary computer programs cannot work.

2.1
History of the Concept of Algorithms

Etymology of the Term "Algorithm"

The word *algorithm* is derived from the name of the mathematician *Muhammad Al-Chwarizmi*, who lived in Baghdad around 780–850. He wrote a book, that was important at the time, on methods for dealing with algebraic equations.

In the Middle Ages, the word *algorismus* was used to describe the art of calculating with Arabic numerals.

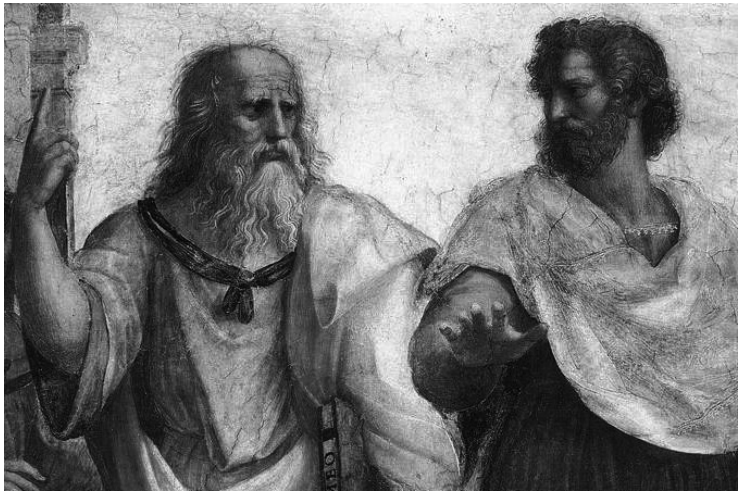
Later, the term *algorithm* was generally used to describe a procedure for solving a problem.

Al-Chwarizmi

Historical Development

The first half of the 20th century saw a rapid development in mathematical logic, which led to the precise definition of the terms *algorithm* and *computability* in the 1930s. The reason for this was not the emergence of computing technology, but the question of the axiomatizability of mathematics:

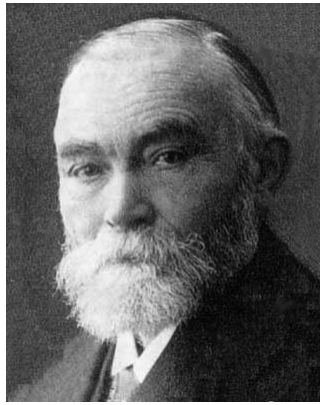
- Even in ancient times, there were discussions about whether there are fundamental limitations to knowledge and predictability. The liar's paradox has been known since the 4th century BC: "This statement is false". The basic ideas behind today's concept of axiomatization can already be found in Plato and Aristotle (400/350 BC).



- Gottfried Leibniz (1646–1716) has a vision of a machine that can manipulate symbols and thus determine the truth value of mathematical statements.

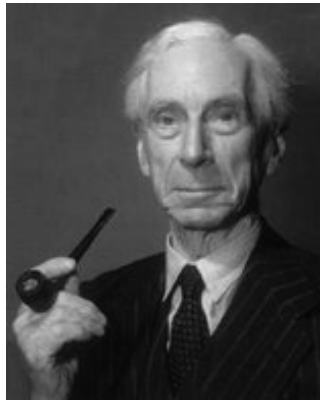
- 1889: Giuseppe Peano (1858–1932) presents the first formal axioms for natural numbers.

- 1893: Gottlob Frege (1848–1925) writes his "Grundgesetze der Arithmetik" (Fundamentals of arithmetic), which are intended to give all of mathematics a purely logical foundation (logicism). All theorems of arithmetic should be traceable to logical truths.



- 1900: David Hilbert (1862–1943) in his speech at the Mathematics Congress in Paris: Paradoxes (like the liar's paradox) in mathematics are based on the ambiguity of natural language. They can be avoided by axiomatization.

- 1902: Bertrand Russel (1872–1970) points out the following paradox to Frege in a letter (*Russel's antinomy*). For $R \stackrel{df}{=} \{x \mid x \notin x\}$ it holds



$$R \in R \Leftrightarrow R \notin R.$$

- Such contradictions lead to the so-called *foundational crisis of mathematics*.

- 1920: Hilbert takes up his idea from 1900 again and develops the so-called Hilbert's program: Call for the complete formalization of mathematics. Theorems should be derived mechanically by applying rules from axioms.

- 1928: Hilbert formulates the decision problem: Is there an algorithm that, given a mathematical statement, determines whether it follows from a given set of axioms?

- 1931: Kurt Gödel (1906–1978) shows the hopelessness of Hilbert's program. *Gödel's incompleteness theorem*: Non-trivial axiomatic systems (e.g. Peano arithmetic) always leave the truth value of certain sentences open.
  "Either mathematics is too big for the human mind or the human mind is more than a machine."

- 1936: Alonzo Church (1903-1995) and Alan Turing (1912-1954) independently show that Hilbert's decision problem cannot be solved by any algorithm. This requires the definition of the terms *algorithm* and *computability*.

  Church uses the lambda calculus, Turing uses the Turing Machine (TM) named after him. It can be seen later that both models have the same computational power.

  Gödel's incompleteness theorem and, above all, the coding of formulas and programs developed by him with natural numbers (Gödelization) had a major influence on the work of Church and Turing.

- Church-Turing thesis (1936): The functions computable in the intuitive sense are exactly the functions computable by TMs. This means that the terms *algorithm* and *computability* can be defined by TMs.
- Further computation models are defined in the following years. It turns out that *all these models are equivalent*. This is how the Church-Turing thesis develops into a generally recognized thesis.
- 1960s: Only now definitions of the term "algorithm" emerge that are characterized by computing technology (e.g. RAMs as a mathematical model for real computers).
- 1970: Extended Church-Turing thesis: Any meaningful computational model can be efficiently simulated (polynomial runtime difference) on a probabilistic TM.

- 1990s: Investigation of new computational models such as quantum computers and DNA computers. It is assumed that certain problems can be solved much faster than in classical computation models.
  However, all problems that can be solved with these computation models can also be solved with classical computation models.

The mathematical characterization of the terms *algorithm* and *computability* is one of the most important scientific achievements of the 20th century.

2.2
Alphabets, Words, Formal Languages

Learning Objectives

After this section you should

1. understand and be able to use the basic terms introduced.
2. understand the dyadic representation.
3. be able to convert safely between decimal notation, binary notation, and dyadic notation.

Computability – The Computer Science Perspective

We have seen that

- the question for precise definitions for terms like "algorithms" and "computation" and
- the question for the limits of these concepts

did not originate from the field of computer technology but from mathematics and the question for axiomatization of mathematical theories.

However, one can also motivate our investigations from a more computer scientific point of view:

What are computers principally capable of and what are their limits?

What are the "things" that computers compute?

They get an input and return an output.
So they compute **functions**.

Input and output are finite sequences of signals which, from a technical point of view, can have a variety of origins: Electricity, electromagnetic waves, quanta, ...

From a theoretical point of view, we summarize all of these in the concept of **words**.

Definition 2.1 (Alphabets and words)

- An **alphabet** is a finite, non-empty set.
  (e.g. $\{0, 1\}$ or $\{a, b, c\}$)
- The elements of an alphabet are called **letters** or **symbols**.
- A **word over an alphabet $\boldsymbol{\Sigma}$** is a finite sequence of 0 or more elements from Σ. (e.g. 00110 or $acbb$)
- We denote the **empty word** (i.e. the word consisting of 0 letters) by ε.

Definition 2.2 (Sets of Words)

Let Σ be an alphabet, $n \geq 0$ and $a_1, a_2, \ldots, a_n \in \Sigma$.

- The **length of a word** $w = a_1 a_2 \cdots a_n$ is $|w| \stackrel{df}{=} n$.
- $\Sigma^n \stackrel{df}{=} \{w \mid w \text{ is a word over } \Sigma \text{ with } |w| = n\}$
  It holds $\Sigma^0 = \{\varepsilon\}$.
- $\Sigma^* \stackrel{df}{=} \{w \mid w \text{ is a word over } \Sigma\} = \bigcup_{n \geq 0} \Sigma^n$ and $\Sigma^+ = \Sigma^* \setminus \{\varepsilon\}$.
- If $v = b_1 b_2 \cdots b_m$ and $w = c_1 c_2 \cdots c_n$, are words over the alphabet Σ, then the **concatenation** of v and w is defined as $\boldsymbol{v \cdot w} \stackrel{df}{=} b_1 b_2 \cdots b_m c_1 c_2 \cdots c_n$.
- A **formal language over** Σ is a subset of Σ^*.
- The **decision problem** of a formal language $L \subseteq \Sigma^*$ is the following task:

  Input: $w \in \Sigma^*$
  Output: 1, if $w \in L$
  0, if $w \notin L$

So, our question

> What are computers principally capable of and what are their limits?

can be reformulated as

> For an alphabet Σ, which functions $\Sigma^* \to \Sigma^*$ can be computed by computers and which cannot?

Before we start modelling computers, we argue that it does not matter whether we consider functions of the form $\Sigma^* \to \Sigma^*$, $\mathbb{N} \to \mathbb{N}$, or $\mathbb{Z} \to \mathbb{Z}$.
We start with binary alphabets.

Binary Representation of Natural Numbers

Property 2.3

Each natural number $n \geq 1$ can be represented in exactly one way as

$$n = \sum_{i=0}^{m} a_i \cdot 2^i$$

with $m \in \mathbb{N}$, $a_m = 1$ and $a_0, \ldots, a_{m-1} \in \{0, 1\}$.

Proof.

Exercise. $\square$

Definition 2.4 (Binary representation)

- The **binary representation** of 0 is 0.
- Let $n \geq 1$ and let $m \in \mathbb{N}$ and $a_0, \ldots, a_m \in \{0, 1\}$ with $a_m = 1$ and $n = \sum_{i=0}^{m} a_i \cdot 2^i$. Then the sequence $a_m a_{m-1} \cdots a_0$ is called **binary representation** of n.
- Let **bin**: $\mathbb{N} \to \{0, 1\}^*$ be defined via $n \mapsto \mathrm{bin}(n)$, where $\mathrm{bin}(n)$ denotes the binary representation of a number $n \in \mathbb{N}$.

Note: By Property 2.3, the function bin is well-defined, i.e., for every natural number there exists a binary representation (existence) and there exists only one (uniqueness). In other words: Property 2.3 shows that bin really is a function and is total.

Example 2.5 (Binary representation)

$\mathrm{bin}(0) = 0$, $\mathrm{bin}(1) = 1$, $\mathrm{bin}(2) = 10$, $\mathrm{bin}(3) = 11$, $\mathrm{bin}(4) = 100$, $\mathrm{bin}(5) = 101$, ...

The next property shows how to compute binary representations.

Property 2.6

The function bin is injective and $bin(2n + a) = bin(n)a$ *for* $n \geq 1$ *and* $a \in \{0, 1\}$.

Proof.

Exercise. $\square$

Example 2.7 (Computing the binary representation)

$\text{bin}(54) = \mathbf{110110}$ since

$$\begin{aligned} 54 &= 2 \cdot 27 + \mathbf{0} \\ 27 &= 2 \cdot 13 + \mathbf{1} \\ 13 &= 2 \cdot 6 + \mathbf{1} \\ 6 &= 2 \cdot 3 + \mathbf{0} \\ 3 &= 2 \cdot 1 + \mathbf{1} \\ 1 &= 2 \cdot 0 + \mathbf{1} \end{aligned}$$

Encoding between Words and Natural Numbers

Recall that by Property 2.6, the binary representation of natural numbers is an injective mapping $\text{bin} : \mathbb{N} \to \{0, 1\}^*$. Its disadvantage: it is not a bijection.

With the dyadic or k-adic representation, we now learn a *bijection* between natural numbers and values.

Property 2.8

Every natural number $n \geq 1$ can be represented in exactly one way as

$$n = \sum_{i=0}^{m} a_i \cdot 2^i$$

with $m \in \mathbb{N}$ and $a_0, \ldots, a_m \in \{1, 2\}$.

Proof.

Exercise. $\square$

Definition 2.9 (Dyadic representation)

$\mathrm{dya} : \mathbb{N} \to \{1,2\}^*$ is defined by

$$\begin{aligned} \mathrm{dya}(0) &\stackrel{df}{=} \varepsilon \\ \mathrm{dya}(n) &\stackrel{df}{=} a_m \cdots a_0, \quad \text{if } n \geq 1,\ n = \textstyle\sum_{i=0}^{m} a_i \cdot 2^i \text{ and } \\ &\qquad\qquad\qquad\quad a_0, \ldots, a_m \in \{1,2\}. \end{aligned}$$

The function dya is well-defined by Property 2.8 and total (existence and uniqueness).

Property 2.10

$dya : \mathbb{N} \to \{1,2\}^*$ is bijective.

Proof.

Follows from Definition 2.9. $\square$

The dyadic representation thus provides a bijection $\mathbb{N} \to \Sigma^*$ for all alphabets Σ with $|\Sigma| = 2$, where the letters from Σ are taken as characters for the digits 1 and 2.

We generalize the representation to bijections $\mathbb{N} \to \Sigma^*$ for all alphabets Σ with $|\Sigma| \geq 2$.

Property 2.11

Let $k \geq 2$. Every natural number $n \geq 1$ can be represented in exactly one way as

$$n = \sum_{i=0}^{m} a_i \cdot k^i$$

with $m \in \mathbb{N}$ and $a_0, \ldots, a_m \in \{1, \ldots, k\}$.

Definition 2.12 (k-adic representation)

For $k \geq 2$ is $\mathrm{ad}_k : \mathbb{N} \to \{1, \ldots, k\}^*$ defined by

$$\begin{aligned} \mathrm{ad}_k(0) &\stackrel{df}{=} \varepsilon \\ \mathrm{ad}_k(n) &\stackrel{df}{=} a_m \cdots a_0, \quad \text{if } n \geq 1,\ n = \textstyle\sum_{i=0}^{m} a_i \cdot k^i \text{ and} \\ & \qquad\qquad\qquad\quad a_0, \ldots, a_m \in \{1, \ldots, k\}. \end{aligned}$$

This means that $\mathrm{dya} = \mathrm{ad}_2$.

Property 2.13

For $k \geq 2$ the function $ad_k : \mathbb{N} \to \{1, \ldots, k\}^*$ is bijective.

The k-adic representation can be determined using the following property.

Property 2.14

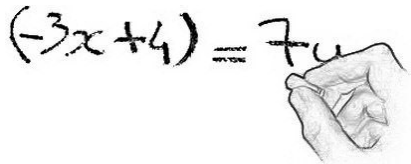
Let $k \geq 2$.

1. $dya(2n + a) = dya(n)a$ *for* $n \geq 0$ *and* $a \in \{1, 2\}$
2. $dya^{-1}(xa) = 2 \cdot dya^{-1}(x) + a$ *for* $x \in \{1, 2\}^*$ *and* $a \in \{1, 2\}$
3. $ad_k(kn + a) = ad_k(n)a$ *for* $n \geq 0$ *and* $a \in \{1, \ldots, k\}$
4. $ad_k{}^{-1}(xa) = k \cdot ad_k{}^{-1}(x) + a$ *for* $x \in \{1, \ldots, k\}^*$, $a \in \{1, \ldots, k\}$

Proof.

Exercise. □

Example 2.15 (Determine the k-adic representation)



If Σ is an alphabet with $k = |\Sigma| \geq 2$, then ad_k yields a bijection $\mathbb{N} \to \Sigma^*$, where the letters from Σ are taken as characters for the digits $1, \ldots, k$.

Therefore, we can now identify natural numbers with words from Σ^* (and thus also words from Σ^* with words form Σ'^* for another alphabet Σ').

In particular, we can use the following encoding $\mathrm{code}_{\Sigma^*} : \Sigma^* \to \mathbb{N}$ to encode words $w \in \Sigma^*$ by natural numbers.

$$\mathbf{code}_{\boldsymbol{\Sigma}^*}(w) \stackrel{df}{=} \mathrm{ad}_k{}^{-1}(w)$$

Encodings between Naturals and Integers

We use the bijection $\text{code}_{\mathbb{Z}} : \mathbb{Z} \to \mathbb{N}$ defined below, to represent integers by natural numbers (and vice versa).

$$\text{code}_{\mathbb{Z}}(x) \stackrel{df}{=} \begin{cases} 2x & \text{if } x \geq 0 \\ -2x - 1 & \text{otherwise} \end{cases}$$

We denote the inverse function of $\text{code}_{\mathbb{Z}}$ by $\text{code}_{\mathbb{Z}}^{-1}$.
We can now identify $\mathbb{Z}$ and $\mathbb{N}$, as one can be encoded by the other.

In order to study the limits of computers, we need to investigate which functions $\mathbb{N} \to \mathbb{N}$ (or alternatively, $\mathbb{Z} \to \mathbb{Z}$ or $\Sigma^* \to \Sigma^*$ for any Σ) can be computed by computers and which cannot.

We now argue that also other types of inputs can be encoded as words, natural numbers, and integers.

Encoding of Lists of Natural Numbers

The input or output of an algorithm is often not just a single number (or word), but a list of numbers (or words). However, the latter can be encoded by *one* number. This means that in many places we can restrict ourselves to algorithms whose inputs or outputs are single numbers. This facilitates our investigations.

We now describe how a list of natural numbers can be encoded by a natural number. Let $n \geq 0$ and $x_1, \ldots, x_n$ be natural numbers with the following binary representations.

$$\begin{aligned}
\mathrm{bin}(x_1) &= a_{1,1}a_{1,2}\cdots a_{1,r_1} \\
\mathrm{bin}(x_2) &= a_{2,1}a_{2,2}\cdots a_{2,r_2} \\
&\vdots \\
\mathrm{bin}(x_n) &= a_{n,1}a_{n,2}\cdots a_{n,r_n}
\end{aligned}$$

The encoding of the list $(x_1, \ldots, x_n)$ is obtained by writing all bits $a_{i,j}$ twice in the above binary representations and appending the resulting strings one after the other, where the bit sequence 10 serves as the start, separator, and end marker. The encoding of the list is thus the natural number with binary representation

$$\begin{aligned} 1\,0\; a_{1,1}\, a_{1,1}\, a_{1,2}\, a_{1,2} \cdots a_{1,r_1}\, a_{1,r_1}\; 1\,0 \\ a_{2,1}\, a_{2,1}\, a_{2,2}\, a_{2,2} \cdots a_{2,r_2}\, a_{2,r_2}\; 1\,0 \\ \vdots \\ a_{n,1}\, a_{n,1}\, a_{n,2}\, a_{n,2} \cdots a_{n,r_n}\, a_{n,r_n}\; 1\,0. \end{aligned}$$

The encoding of the list $(x_1, \ldots, x_n)$ is denoted by $\langle x_1, \ldots, x_n \rangle$.

Example: $\langle\rangle = \mathrm{bin}^{-1}(10) = 2$

$\langle 2 \rangle = \mathrm{bin}^{-1}(10110010) = 178$

$\langle 5, 3, 2 \rangle = \mathrm{bin}^{-1}(1011001110111110110010) = 2944946$

Encoding of Lists of Other Objects

Algorithms not only process natural numbers but also other objects such as integers or words. For objects of type $\mathcal{X}$ suitable encodings

$$\mathrm{code}_{\mathcal{X}} : \mathcal{X} \to \mathbb{N}$$

will be defined. This allows us to encode a list $(x_1, \ldots, x_n)$ of objects of type $x_1 \in \mathcal{X}_1, \ldots, x_n \in \mathcal{X}_n$ by the natural number

$$\langle \mathrm{code}_{\mathcal{X}_1}(x_1), \ldots, \mathrm{code}_{\mathcal{X}_n}(x_n) \rangle.$$

If the types of the objects $x_1, \ldots, x_n$ are clear, we write in abbreviated form

$$\langle x_1, \ldots, x_n \rangle.$$

Having clarified that it does not matter which kind of inputs we consider, we start modelling computers.

2.3
While Programs

Learning Objectives

After this section you should

1. understand inductive definitions.
2. know the syntax, semantics and working mechanism of While and Loop programs.
3. understand and be able to apply the concept of While-computability in detail.
4. be able to create simple While programs yourselves and be able to determine the computed function for simple While programs.
5. know the relationship between LOOP and WHILE and understand the role of the Ackermann function.

Programming Languages

Programming languages not only make programming easier, but also offer an *abstraction* of real computers. This means that once a program has been written, it can be transferred to other hardware. A programming language is therefore a mathematical model of a computer, a so-called *computation model*.

The list of known programming languages is long and new ones are added every year.

List of Programming Languages

Wikipedia list of programming languages (as of 07/2024):A.NET, A-0 System, A+, ABAP, ABC, ABC ALGOL, ACC, Accent, Ace Distributed Application Specification Language, Action!, ActionScript, Actor, Ada – ISO/IEC 8652, Adenine, AdvPL, Agda, Agilent VEE, Agora, AIMMS, Aldor, Alef, ALF, ALGOL 58, ALGOL 60, ALGOL 68, ALGOL W, Alice ML, Alma-0, AmbientTalk, Amiga E, AMPL, Analitik, AngelScript, Apache Pig latin, Apex, APL, App Inventor for Androids visual block language, AppleScript, APT, Arc, ArkTS, ARexx, Argus, Assembly language, AutoHotkey, AutoIt, AutoLISP / Visual LISP, Averest, AWK, Axum, B, Babbage, Ballerina, Bash, BASIC, Batch file, bc, BCPL, BeanShell, BETA, BLISS, Blockly, BlooP, Boo, Boomerang, Bosque, C , C–, C++, C*, C#, C/AL, Caché ObjectScript, C Shell, Caml, Cangjie, Carbon, Catrobat, Cayenne, Cecil, CESIL, Céu, Ceylon, CFEngine, Cg, Ch, Chapel, Charm, CHILL, CHIP-8, ChucK, Cilk, Claire, Clarion, Clean, Clipper, CLIPS, CLIST, Clojure, CLU, CMS-2, COBOL, CobolScript – COBOL Scripting language, Cobra, CoffeeScript, ColdFusion, COMAL, COMIT, Common Intermediate Language, Common Lisp, COMPASS, Component Pascal, COMTRAN, Concurrent Pascal, Constraint Handling Rules, Control Language, Coq, CORAL, Coral 66, CorVision, COWSEL, CPL, Cryptol, Crystal, Csound, Cuneiform, Curl, Curry, Cybil, Cyclone, Cypher Query Language, Cython, CEEMAC, D, Dart, Darwin, DataFlex, Datalog, DATATRIEVE, dBase, dc, DCL, Delphi, DIBOL, DinkC, Dog, Draco, DRAKON, Dylan, DYNAMO, DAX, E, Ease, Easy PL/I, EASYTRIEVE PLUS, ECMAScript, Edinburgh IMP, EGL, Eiffel, ELAN, Elixir, Elm, Emacs Lisp, Emerald, Epigram, EPL, Erlang, es, Escher, ESPOL, Esterel, Etoys, Euclid, Euler, Euphoria, EusLisp Robot Programming Language, CMS EXEC, EXEC 2, Executable UML, Ezhil, F, F#, F*, Factor, Fantom, FAUST, FFP, fish, Fjölnir, FL, Flavors, Flex, Flix, FlooP, FLOW-MATIC, FOCAL, FOCUS, FOIL, FORMAC, @Formula, Forth, Fortran – ISO/IEC 1539, Fortress, FP, FoxBase/FoxPro, Franz Lisp, Futhark, Game Maker Language, GameMonkey Script, General Algebraic Modeling System, GAP, G-code, GDScript, Genie, Geometric Description Language, GEORGE, OpenGL Shading Language, GNU E, GNU Ubiquitous Intelligent Language for Extensions, Go, Go!, Game Oriented Assembly Lisp, Gödel, Golo, Good Old Mad, Google Apps Script, Gosu, GOTRAN, General Purpose Simulation System, GraphTalk, GRASS, Grasshopper, Groovy, Hack, HAGGIS, HAL/S, Halide, Hamilton C shell, Harbour, Hartmann pipelines, Haskell, Haxe, Hermes, High Level Assembly, High Level Shader Language, Hollywood, HolyC, Hop, Hopscotch, Hope, Hume, HyperTalk, Hy, Io, Icon, IBM Basic assembly language, IBM Informix-4GL, IBM RPG, IDL, Idris, Inform, ISLISP, J, J#, J++, JADE, Jai, JAL, Janus, Janus, JASS, Java, JavaFX Script, JavaScript, Jess, JCL, JEAN, Join Java, JOSS, Joule, JOVIAL, Joy, jq, JScript, JScript .NET, Julia, Jython, K, Kaleidoscope, Karel, KEE, Kixtart, Klerer-May System, KIF, Kojo, Kotlin, KRC, KRL, KRL, KRYPTON, KornShell, Kodu, Kv, LabVIEW, Ladder, LANSA, Lasso, Lava, LC-3, Lean, Legoscript, LIL, LilyPond, Limbo, LINC, Lingo, LINQ, LIS, LISA, Language H, Lisp – ISO/IEC 13816, Lithe, Little b, LLL, Logo, Logtalk, LotusScript, LPC, LSE, LSL, LiveCode, LiveScript, Lua, Lucid, Lustre, LYaPAS, Lynx, M, M4, M#, Machine code, MAD, MAD/I, Magik, Magma, Maple, MAPPER, MARK-IV, Mary, MATLAB, MASM Microsoft Assembly x86, MATH-MATIC, Maude system, Maxima, Max, MaxScript internal language 3D Studio Max, Maya, MDL, Mercury, Mesa, MHEG-5, Microcode, Microsoft Power Fx, MIIS, MIMIC, Mirah, Miranda, MIVA Script, ML, Model 204, Modelica, Malbolge, Modula, Modula-2, Modula-3, Mohol, Mojo, MOO, Mortran, Mouse, MPD, MSL, MUMPS, MuPAD, Mutan, Mystic Programming Language, NASM, Napier88, Neko, Nemerle, NESL, Net.Data, NetLogo, NetRexx, NewLISP, NEWP, Newspeak, NewtonScript, Nial, Nickle, Nim, Nix, NPL, Not eXactly C, Not Quite C, NSIS, Nu, NWScript, NXT-G, o:XML, Oak, Oberon, OBJ2, Object Lisp, ObjectLOGO, Object REXX, Object Pascal, Objective-C, Obliq, OCaml, occam, occam-π, Octave, OmniMark, Opa, Opal, Open Programming Language, OpenCL, OpenEdge Advanced Business Language, OpenVera, OpenQASM, OPS5, OptimJ, Orc, ORCA/Modula-2, Oriel, Orwell, Oxygene, Oz, P, P4, P", ParaSail, PARI/GP, Pascal – ISO 7185, Pascal Script, PCASTL, PCF, PEARL, PeopleCode, Perl, PDL, Pharo, PHP, Pico, Picolisp, Pict, Pike, PILOT, Pipelines, Pizza, PL-11, PL/0, PL/B, PL/C, PL/I – ISO 6160, PL/M, PL/P, PL/S, PL/SQL, PL360, PLANC, Plankalkül, Planner, PLEX, PLEXIL, Plus, POP-11, POP-2, PostScript, PortablE, POV-Ray SDL, Powerhouse, PowerBuilder – 4GL GUI application generator from Sybase, PowerShell, PPL, Processing, Processing.js, Prograph, Project Verona, Prolog, PROMAL, Promela, PROSE modeling language, PROTEL, Pro*C, Pure, Pure Data, PureScript, PWCT, Python, Q, Q#, Qalb, Quantum Computation Language, QtScript, QuakeC, QPL, .QL, R, R++, Racket, Raku, RAPID, Rapira, Ratfiv, Ratfor, rc, Reason, REBOL, Red, Redcode, REFAL, REXX, Ring, ROOP, RPG, RPL, RSL, RTL/2, Ruby, Rust, S, S2, S3, S-Lang, S-PLUS, SA-C, SabreTalk, SAIL, SAKO, SAS, SASL, Sather, Sawzall, Scala, Scheme, Scilab, Scratch, ScratchJr, Script.NET, Sed, Seed7, Self, SenseTalk, SequenceL, Serpent, SETL, Short Code, SIMPOL, SIGNAL, SiMPLE, SIMSCRIPT, Simula, Simulink, SISAL, SKILL, SLIP, SMALL, Smalltalk, SML, Strongtalk, Snap!, SNOBOL, Snowball, SOL, Solidity, SOPHAEROS, Source, SPARK, Speakeasy, Speedcode, SPIN, SP/k, SPL, SPS, SQL, SQR, Squeak, Squirrel, SR, S/SL, Starlogo, Strand, Stata, Stateflow, Subtext, SBL, SuperCollider, Superplan, SuperTalk, Swift, Swift, SYMPL, T, TACL, TADS, TAL, Tcl, Tea, TECO, TELCOMP, TeX, TEX, TIE, TMG, compiler-compiler, Tom, Toi, Topspeed, TPU, Trac, TTM, T-SQL, Transcript, TTCN, Turing, TUTOR, TXL, TypeScript, Tynker, Ubercode, UCSD Pascal, Umple, Unicon, Uniface, UNITY, UnrealScript, V, Vala, Verse, Vim script, Viper, Visual DataFlex, Visual DialogScript, Visual FoxPro, Visual J++, Visual LISP, Visual Objects, Visual Prolog, WATFIV, WATFOR, WebAssembly, WebDNA, Whiley, Winbatch, Wolfram Language, Wyvern, X++, X10, xBase++, XBL, XC, xHarbour, XL, Xojo, XOTcl, Xod, XPL, XPL0, XQuery, XSB, XSLT, Xtend, Yorick, YQL, Z++, Z shell, Zebra, ZPL, ZPL2, Zeno, ZetaLisp, Zig, Zonnon, ZOPL, ZPL.

We will study the computation models based on programming languages on the basis of While programs, whose syntax is based on the Python programming language.
In principle, we could also choose another programming language, and we would get the same results. The advantage of While programs is that they only have a limited syntax. This makes it easier for us to carry out mathematical proofs later on.

Programming languages are usually defined in two stages:

1. The **definition of the syntax** determines *which* language element is used and *how*. The syntax clearly determines the set of syntactically correct programs.
2. The **definition of semantics** determines the meaning/way of using the language elements. Semantics thus describes the behavior of syntactically correct programs.

At the latest when programming the compiler, it becomes clear that the precise definition of a programming language is necessary in order to automatically translate source code into machine code.

To define the syntax and semantics of While programs we need the following logical connectors.

Definition 2.16 (logical connectors)

Let a and b be truth values (i.e., true=1 and false=0).

$$\neg a \stackrel{df}{=} \begin{cases} \text{true}, & \text{if } a = \text{false} \\ \text{false} & \text{else} \end{cases}$$

$$a \wedge b \stackrel{df}{=} \begin{cases} \text{true}, & \text{if } a = b = \text{true} \\ \text{false} & \text{else} \end{cases}$$

$$a \vee b \stackrel{df}{=} \begin{cases} \text{true}, & \text{if } a = \text{true or } b = \text{true} \\ \text{false} & \text{else} \end{cases}$$

Syntax and Semantics of While Programs

The syntax defines the following language elements: Identifiers, constants, variables, expressions, conditions, statement blocks, function declarations, While programs.
For a better understanding, we define the syntax and semantics of While programs in parallel.

The semantics are defined in boxes like this one.

1. Identifiers:

An identifier is a finite sequence of letters `a`,...,`z`,`A`,...,`Z` and digits `0`,...,`9` that starts with a letter and is none of the following words:

`not, and, or, if, else, while, for, in, range, def, return, print, False, class, finally, is, None, continue, lambda, as, try, True, from, nonlocal, del, global, with, elif, yield, assert, async, await, import, pass, break, except, raise`

- Differentiation between upper and lower case letters
- Used for variable and function names

Examples: `a, b3, ab15cd, house, hOuse`

The reserved identifiers are Python keywords. The words `not`,...,`print` are keywords of While programs.

2. Constants:

- `0` and `-0` are constants.
- If *a* is a non-empty, finite sequence of digits `0`, . . . , `9` that does not start with `0`, then *a* and `-`*a* are constants.

Constants describe integers in decimal representation.

Examples: `6362, -53, 0, -0`

3. Variables:

If *a* is an identifier, then *a* is a variable.

The value of a variable is not defined until the first assignment. Afterwards the variable describes an integer (which is also called the assignment of the variable).

Examples: `a, b3, ab15cd, house, hOuse`

4. **Expressions:** (Inductive Definition)

An expression describes an integer (if it is defined).

BC **Basic expressions**

- Each constant and each variable is an expression.

The integer described by the constant or variable.

IS **Compound expressions**

- If a and b are expressions, so are `(`a `+` b`)` and `(`a `-` b`)`.

a, b defined $\Rightarrow$ expressions `(`a `+` b`)`, `(`a `-` b`)` stand for the integers $a+b$, $a-b$.
a or b not defined $\Rightarrow$ `(`a `+` b`)`, `(`a `-` b`)` not defined.

- If f is an identifier, $n \geq 0$, and $b_1, \ldots, b_n$ are expressions, then f`(`b_1`,...,`b_n`)` is an expression (**function call**).

The semantics will be defined under "Function declarations".

Examples: `w5, 7, f(b), f(), h((a+b)), (f((3+g(9)),2)-7)`

5. **Conditions:** (Inductive definition)

BC If a, b are expressions, then `(a < b)`, `(a <= b)`, `(a > b)`, `(a >= b)`, `(a == b)`, `(a != b)` are conditions.

a, b defined $\Rightarrow$ the conditions are assigned the usual truth values *true* or *false*.
a or b not defined $\Rightarrow$ conditions not defined.

IS If a, b conditions, so are `(not a)`, `(a and b)` and `(a or b)`.

a, b defined $\Rightarrow$ the conditions have the truth values $\neg a$, $a \wedge b$, $a \vee b$.
a not defined $\Rightarrow$ (`not` a), (a `and` b), (a `or` b) not defined.
a true, b not defined $\Rightarrow$ (a `or` b) true, (a `and` b) not defined.
a false, b not defined $\Rightarrow$ (a `and` b) false, (a `or` b) not defined

Examples: `(0 == 1)`, `((5 - a) < f(3))`, `(not (a < f(3)))`

6. **Instruction blocks:** (Inductive definition)

BC **Assignments**

- If a is a variable and b an expression, then the following is an instruction block.

  $a = b$

b defined $\Rightarrow$ the value of b is assigned to the variable a
b not defined $\Rightarrow$ computation of instruction block undefined.

Examples: `a = 2, a = b, a = ((d + e) - f(10))`

IS Compound instruction blocks

- **Consecutive execution**: If $n \geq 2$ and $s_1, \ldots, s_n$ are instruction blocks, then the following is an instruction block.

  s_1
  s_2
  $\vdots$
  s_n

  ($s_1, \ldots, s_n$ are placed one below the other with the same indentation)

  > The instruction blocks are executed in the order $s_1, \ldots, s_n$. If one of the s_i is not defined, then the computation of the complete instruction block is not defined.

  Example:

  ```
  a = (a - 1000000000000000000000000000000)
  b = f(a)
  c = ((b + b) - a)
  ```

- **If-instructions**: For a condition b and instruction blocks s_1, s_2 the following two constructs are also instruction blocks:

```
if b:
  s1
else:
  s2
```

```
if b:
  s1
```

(s_1 and s_2 are indented further than `if` and `else`)

`if` b`:`
s_1
`else:`
s_2
	b true $\Rightarrow$ s_1 is executed.
	b false $\Rightarrow$ s_2 is executed.
	b not defined $\Rightarrow$ coputation not defined

`if` b`:` s_1	b true $\Rightarrow$ s_1 is executed. b false $\Rightarrow$ next instruction is executed. b not defined $\Rightarrow$ computation not defined

Example:

```
a = 0
if (a <= a):
  a = (a - 10)
  b = f(a)
  c = ((b + b) - a)
if (1 == 1):
  a = 1
else:
  a = 2
a = 0
```

- **While-loops**: For a condition b and an instruction block s the following is an instruction block.

  ```
  while b:
    s
  ```

  (s is indented further than `while`)

  > The condition b is evaluated first. If it is not defined, the computation of the while-loop is not defined. If b is false, the while-loop is ended and the next instruction is executed. If b is true, s is executed first and then the while-loop is started again (with the evaluation of the condition b). The instruction block s is therefore repeated as long as b is true. If b is never false, the computation of the while-loop is not defined.

  Example:

  ```
  while (a != 0):
    a = (a - 1)
    b = (b + a)
  ```

- **For-loops**: For a variable i, expressions a_1, a_2 and an instruction block s the following is an instruction block:

  ```
  for i in range(a1,a2):
    s
  ```

  (s is indented further than `for`)

  The working mechanism corresponds to the following instruction block, where j,k are two variables not used in the rest of the program.

  ```
  j = a1
  k = a2
  while (j < k):
    i = j
    s
    j = (j + 1)
  ```

  Example:

  ```
  for i in range(1,3):
    i = 1
  ```

7. Function declarations:

- **Function declaration without initialization:** Let f be an identifier, $n \geq 0$, $a_1, \ldots, a_n$ pairwise distinct variables, s an instruction block and c an expression. Then the following two constructs are declarations of the function f.

```
def f(a1,...,an):
  s
  return c
```

```
def f(a1,...,an):
  return c
```

Semantics of the function call $f(c_1, \ldots, c_n)$, where $c_1, \ldots, c_n$ are expressions: If one of the expressions $c_1, \ldots, c_n$ is not defined, the computation of the function call is not defined. Otherwise, s (if available) is executed, whereby the variables $a_1, \ldots, a_n$ are assigned the values $c_1, \ldots, c_n$ and the values of the remaining variables are not defined.[7] If this computation is defined, the value of the function call $f(c_1, \ldots, c_n)$ is is defined as the value of c (at the end of the computation). Otherwise, the computation of the function call is not defined.

[7] Local variable visibility

- **Function declaration with initialization:** Let f, n, $a_1, \ldots, a_n$, s, c be as above. Let $b_1, \ldots, b_m$ with $m \geq 0$ the variables used in s and c without $a_1, \ldots, a_n$. Then the following two constructs are declarations of the function f.

```
def f(a1,...,an):
  [b1,...,bm] = [0,...,0]
  s
  return c
```

```
def f(a1,...,an):
  [b1,...,bm] = [0,...,0]
  return c
```

The above lists `[0,...,0]` contain m zeros.

The semantics correspond to the following function declarations:

```
def f(a1,...,an):
  b1 = 0
   ⋮
  bm = 0
  s
  return c
```

```
def f(a1,...,an):
  b1 = 0
   ⋮
  bm = 0
  return c
```

Examples:

```
def f(x,y):
  z = 0
  while (x > 0):
    z = (z + y)
    x = (x - 1)
  return z

def g(x,y):
  [i,j,z] = [0,0,0]
  for i in range(x,y):
    j = (j + i)
  z = (j + j)
  return z

def h(x,y):
  [] = []
  return (x + y)
```

8. While-programs:

A While-Program is a finite non-empty sequence of function declarations with the following properties:

1. Each function called in the program is declared with the appropriate arity.
2. If `def` f`(...):` and `def` g`(...):` are two function declarations, the identifiers f and g are distinct.
3. If `def` f`(...):` is a function declaration and v a variable of the program, then the identifiers f and v are distinct.

If

```
def f(a_1,...,a_n):
    ...
```

is the last function declaration in a While program P, then **the function computed by P** is the function $\psi_P : \mathbb{Z}^n \to \mathbb{Z}$ with:

$$\psi_P(x_1, \ldots, x_n) \stackrel{df}{=} \begin{cases} f(x_1, \ldots, x_n), & \text{if this is defined} \\ \text{not defined} & \text{else} \end{cases}$$

Visibility of variables

The semantics of function calls are defined in such a way that variables are only visible in the corresponding function.

Example:

```
def h(x):
  if (0 == 0):
    a = 0
    if (x == 0):
      b = 0
      c = g(0)
  return a     # no error because a was defined

def g(y):
  z = b        # program termination as b not defined
  return z
```

Definition 2.17 (While computability)

- An **algorithm** is a While program.
- A function $\varphi : \mathbb{Z}^n \to \mathbb{Z}$ with $n \geq 0$ is called **While computable** if there is a While program P such that $D_{\psi_P} = D_\varphi$ and $\psi_P(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
- A function $\varphi : \mathbb{N}^n \to \mathbb{N}$ with $n \geq 0$ is called **While computable** if there is a While program P such that $D_{\psi_P} \cap \mathbb{N}^n = D_\varphi$ and $\psi_P(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
- **WHILE** $\stackrel{df}{=} \{\varphi \mid \varphi : \mathbb{N}^n \to \mathbb{N}$ is While computable$\}$.

We thus have our first mathematical characterization of the terms *algorithm* and *computability*.

Self calls of functions

We name the call of a function within its own declaration **self-call** or also **recursion**.

The possibility of self-calls or mutual calls of several functions is called **recursive programming**.

Example:

```
def f(x):
  y = 1
  if (x > 0):
    y = (f((x - 1)) + f((x - 1)))
  return y
```

```
def f(x):
  y = 1
  if (x > 0):
    y = (f((x - 1)) + f((x - 1)))
  return y
```

By the definition of the program, it computes the function $g : \mathbb{Z} \to \mathbb{Z}$ defined via

$$x \mapsto \begin{cases} 1 & \text{if } x \leq 0 \\ 2 \cdot f(x-1) & \text{else.} \end{cases}$$

We now prove that for all $n \in \mathbb{N}$ it holds $g(n) = 2^n$ via induction.

Proof.

BC: For $k = 0$ it holds $2^k = 1$ and $g(k) = 1$.
IS: Let's assume that for some $k \in \mathbb{N}$ it holds $g(k) = 2^k$. We have to prove that then also $g(k+1) = 2^{k+1}$.
This is so: $g(k+1) = 2 \cdot g(k) \overset{\text{ind. hypothesis}}{=} 2 \cdot 2^k = 2^{k+1}$. $\square$

Causes of undefinedness in While-computable functions:

- endless loops, e.g.:

```
def f(x):
  while (x != 0):
    x = (x - 1)
  return x
```

  $\Rightarrow$ $f(x) = 0$ for $x \geq 0$ and $f(x)$ not defined for $x < 0$.

- certain recursive calls, e.g.:

```
def f(x):
  x = f((x + 1))
  return x
```

  $\Rightarrow$ $f(x)$ not defined for all $x \in \mathbb{Z}$.

- access to undefined variables, e.g.:

```
def f(x):
  return z
```

  $\Rightarrow$ $f(x)$ not defined for all $x \in \mathbb{Z}$.

We eliminate these causes in the following computation model.

Definition 2.18 (Loop program)

A **Loop program** is a While program with the following properties:

1. The program does not contain any while-loops.
2. Only functions that are declared above can be called by each function. In particular, no self-calls are permitted.
3. The program only contains function declarations with initialization.

Definition 2.19 (Loop computability)

- A function $\varphi : \mathbb{Z}^n \to \mathbb{Z}$ with $n \geq 0$ is called **Loop-computable** if there is a Loop program P such that $D_{\psi_P} = D_\varphi$ and $\psi_P(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
- A function $\varphi : \mathbb{N}^n \to \mathbb{N}$ with $n \geq 0$ is called **Loop-computable** if there is a loop program P such that $D_{\psi_P} \cap \mathbb{N}^n = D_\varphi$ and $\psi_P(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
- **LOOP** $\stackrel{df}{=} \{\varphi \mid \varphi : \mathbb{N}^n \to \mathbb{N}$ is Loop-computable$\}$.

Special constructs for While and Loop programs

In order to simplify working with While and Loop programs we define in addition to the previously defined syntax the following "comfort elements":

- If $a_1, \ldots, a_n$ are expressions with $n \geq 1$, then `print(`$a_1, \ldots, a_n$`)` is a statement block. It outputs the values of expressions to the screen for testing purposes.
- Comments are marked with #, i.e., the text between # and the end of the line is ignored.

Example:

```
def f(a,b):          # computes the sum of two numbers
  print(a,b)         # outputs the values of a and b
  return (a + b)     # return the result
```

Notes on working with While and Loop programs

- We have defined While and Loop programs in such a way that they form sublanguages of the Python 3 programming language. In particular, While and Loop programs can be executed directly in Python 3.
- You will find a syntax checker online that checks whether a given source code is a While or Loop program.
- In addition to the syntax of While and Loop programs, we also allow the insertion/omission of spaces, provided that this does not change the semantics of the program. (e.g. `a=(b+c)` instead of `a = (b + c)`)
- Do not use tabs and avoid empty lines in functions (use lines with # instead).
- When accessing undefined expressions, Python sometimes throws exceptions.

Example 2.20

Loop program for the multiplication in $\mathbb{Z}$.

```
def prodZ(x,y):
  [i,z] = [0,0]            # Initialization
  if (x < 0):
    x = (0 - x)            # remove negative sign from x
    y = (0 - y)            # and transfer to y
  for i in range(0,x):     # x loop executions
    z = (z + y)            # y is added x times to z
  return z
```

For the input $x = -3$ and $y = 12$, we note the situation after each line. This results in the following flow chart.

Action	x	y	i	z	Comment
	-3	12	n.d.	n.d.	
`[i,z] = [0,0]`	-3	12	0	0	
`if (x < 0):`	-3	12	0	0	true
`x = (0 - x)`	3	12	0	0	
`y = (0 - y)`	3	-12	0	0	
`for i in range(0,x):`	3	-12	0	0	i = 0
`z = (z + y)`	3	-12	0	-12	
`for i in range(0,x):`	3	-12	1	-12	i = 1
`z = (z + y)`	3	-12	1	-24	
`for i in range(0,x):`	3	-12	2	-24	i = 2
`z = (z + y)`	3	-12	2	-36	
`for i in range(0,x):`	3	-12	2	-36	end of loop
`return z`	3	-12	2	-36	return -36

Example 2.21

While program for the rounding off division in $\mathbb{Z}$.

$$\mathrm{divZ}(x, y) \stackrel{df}{=} \begin{cases} \lfloor x/y \rfloor, & \text{if } y \neq 0 \\ \text{n.d.}, & \text{if } y = 0, \end{cases}$$

where for $y \neq 0$

$$\lfloor x/y \rfloor \stackrel{df}{=} \begin{cases} \text{largest } z \in \mathbb{Z} \text{ with } y \cdot z \leq x, & \text{if } y > 0 \\ \lfloor (-x)/(-y) \rfloor, & \text{if } y < 0 \end{cases}$$

```
def divZ(x,y):
  if (y != 0):       # return undefined if y == 0
    if (y < 0):
      y = (0 - y)    # remove neg. sign from y
      x = (0 - x)    # and transfer to x
    z = 0
    if (x < 0):      # z = min(0,x)
      z = x
                     # here y>0 and y*z <= x
    while (prodZ(y,z) <= x): # find max z with y*z<=x
      z = (z + 1)
    z = (z - 1)      # correction, counted too far
  return z
```

Example 2.22

A While program with several function declarations. Three functions are declared in the program, which are defined for $n, m, i \in \mathbb{Z}$ as follows.

$$\begin{aligned}
\text{modZ}(x, y) &\stackrel{df}{=} x - (y \cdot \text{d}ivZ(x, y)) \\
\text{divisors}(x) &\stackrel{df}{=} \begin{cases} \text{number of divisors of } x, \text{ if } x \geq 1 \\ 0, \text{ else} \end{cases} \\
\text{prime}(n) &\stackrel{df}{=} \begin{cases} n\text{-th prime number, if } n \geq 0 \\ 2, \text{ else} \end{cases}
\end{aligned}$$

For $x \in \mathbb{N}$ and $y \in \mathbb{N}^+$, $modZ(x, y)$ is the remainder of the division x/y.
In the definition of prime, we consider 2 to be the 0th prime.
The While program itself computes the function prime.

```
def modZ(x,y):
  return (x - prodZ(y,divZ(x,y)))

def divisors(x):
  k = 0                           # set counter to 0
  for i in range(1,(x + 1)):      # i = 1, ..., x
    if (modZ(x,i) == 0):
      k = (k + 1)                 # increment counter
  return k

def prim(n):
  k = 0
  m = 2                  # 0-th prime number
  while (n > k):         # in {2,3,...,m} there are the
    m = (m + 1)          # 0-th, 1-st, ..., k-th prime
    if (divisors(m) == 2): # (k+1)-th prime found
      k = (k + 1)
  return m
```

Example 2.23 (While Programs for the Factorial)

```
def fac(x):          # fac(x) = x!, if x>=1
  y = 1              # fac(x) = 1, otherwise
  for i in range(1,(x + 1)):
    y = prodZ(i,y)
  return y

def fac2(x):                        # fac2(x) = x!, if x>=1
  if (x > 1):                       # fac2(x) = x, otherwise
    x = prodZ(x,fac2((x - 1))) # Self-call of fac2
  return x

def fac3(x):       # fac3(x) = x!, if x>=1
  y = 1            # fac3(x) = n.d., otherwise
  while (x != 1):
    y = prodZ(x,y)
    x = (x - 1)
  return y
```

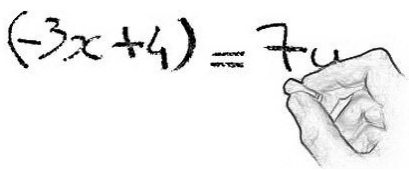
$\Rightarrow$ 3 different functions are computed

Example 2.24

Which functions are computed by the following While programs?

```
def f1(x,y):
  z = 0
  while (x > y):
    x = (x + 2)
    y = (y + 3)
    z = (z + 1)
  return z
```

```
def f2(x):
  while (x>1):
    z=x
    x=0
    while (z>1):
      z=(z-2)
      x=(x+1)
    if (z>0):
      x=((x+x)+x)
      x=((x+x)+4)
  return 0
```



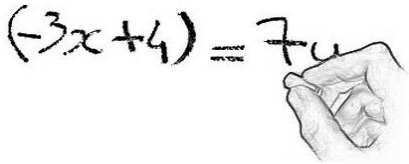
Example 2.25

Which functions are computed by the following Loop programs?

```
def f3(x):
  [y,i]=[0,0]
  for i in range (0,x):
    y = (y + x)
  return y
```

```
def f4(x):
  [i]=[0]
  for i in range (0,x):
    x = (x + 2)
  return x
```

```
def f5(x):
  [i]=[0]
  for i in range (0,x):
    x = (x + x)
  return x
```



Theorem 2.26 (Loop vs While computability)

1. *Every Loop-computable function* $\varphi : \mathbb{Z}^n \to \mathbb{Z}$ *with* $n \geq 0$ *is While-computable.*
2. *Every Loop-computable function* $\varphi : \mathbb{N}^n \to \mathbb{N}$ *with* $n \geq 0$ *is While-computable.*
3. LOOP $\subsetneq$ WHILE.

Proof idea:

- Loop programs are special While programs
- Strictness of inclusion: Loop programs are always total, but While programs are not

Proof.

Since Loop programs are special While programs, statements 1 and 2 and LOOP $\subseteq$ WHILE hold.

Loop programs do not contain infinite loops and infinite cycles of function calls are not possible in Loop programs. It is also not possible to access undefined variables.

This means that Loop programs halt on every input with a defined result. So all Loop-computable functions $\mathbb{N}^n \to \mathbb{N}$ are total.

Consequently, $g : \mathbb{N} \to \mathbb{N}$ with $g(x) \stackrel{df}{=}$ n.d. is not Loop-computable.

However, the function g is While-computable:

```
def g(x):
  while (0 == 0):
    x = 0
  return 0
```

Thus $g \in \text{WHILE} - \text{LOOP}$ and therefore $\text{LOOP} \neq \text{WHILE}$. $\square$

Ackermann Function

Theorem 2.26 raises the following question:

Does $\text{LOOP} = \{f \in \text{WHILE} \mid f \text{ is total}\}$ hold?

In 1926 Wilhelm Ackermann proved that the following function $a : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ (also called **Ackermann function**) is total and While-computable, but is not Loop-computable.

$$a(n, m) \stackrel{df}{=} \begin{cases} m + 1 & \text{if } n = 0 \\ a(n-1, 1) & \text{if } n > 0 \text{ and } m = 0 \\ a(n-1, a(n, m-1)) & \text{if } n > 0 \text{ and } m > 0 \end{cases}$$

Theorem 2.27

The Ackermann function is a total function in $\text{WHILE} - \text{LOOP}$.

Without proof.

The Ackermann function grows extremely quickly:

$$\begin{aligned} a(1,3) &= 5 \\ a(2,3) &= 9 \\ a(3,3) &= 61 \\ a(4,3) &\approx 10^{10^{19728}} \end{aligned}$$

2.4
Register Machines

Learning Objectives

After this section you should

1. understand the structure and working mechanism of RAMs.
2. know the RAM instructions and their semantics.
3. understand and be able to apply the concept of RAM computability in detail.
4. be able to create simple RAM programs yourselves and be able to determine the computed function for simple RAM programs.
5. know the differences between While and Python programs.
6. be able to outline the translation of RAMs into While programs.
7. understand the idea of dynamic programming.

We want to develop a mathematical model for *computers*.

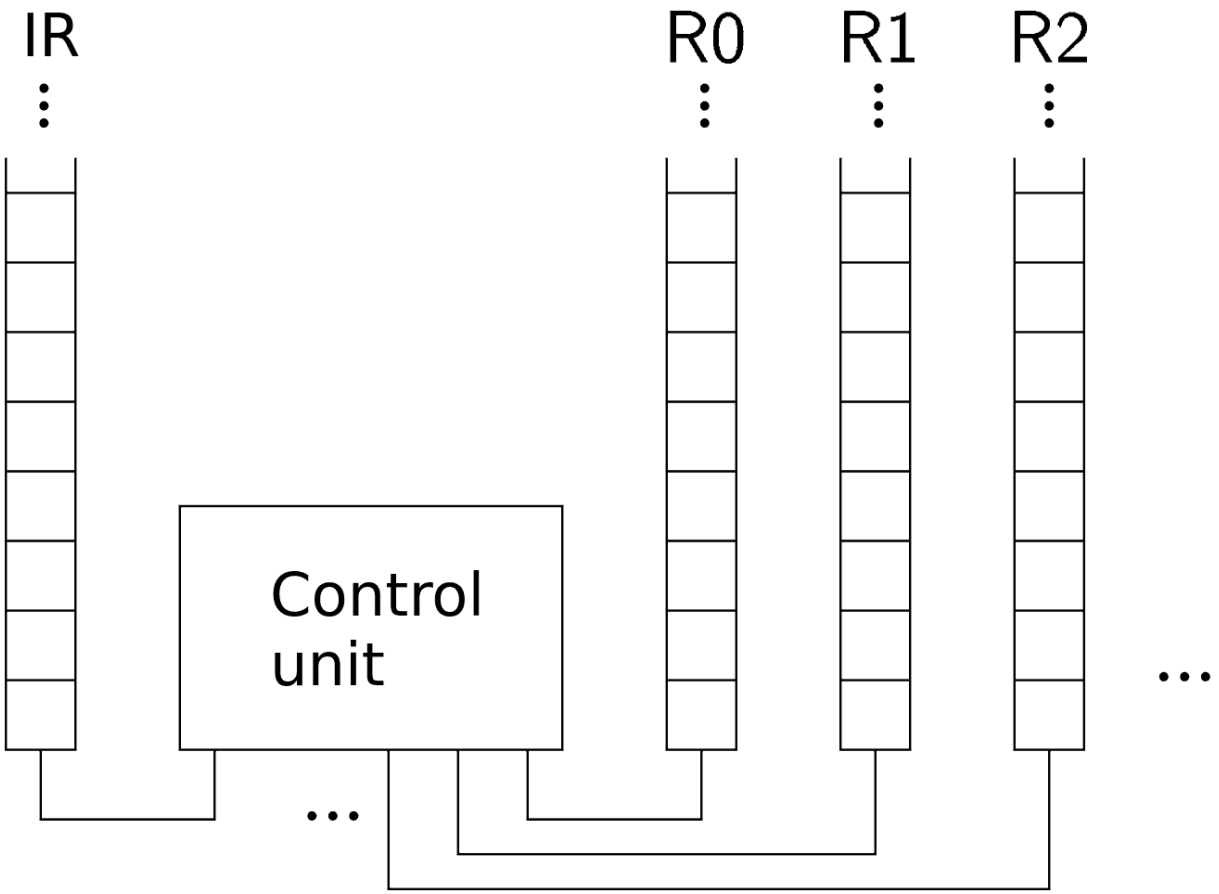
Reason: real computers are very different in terms of

- Processor
- Architecture
- Computing power
- Memory capacity
- ...

Advantage of mathematical models:
Abstraction of technical details leads to better understanding.

With *random access machines (RAM)* we get to know another computation model. It captures the essential properties of real computers, but abstracts from technical details.

Structure of a RAM:



- each of the registers IR, R0, R1, ... contains a natural number
- the control unit has a program
  (finite list of instructions, numbered with $0, 1, 2, \ldots$) .
- different RAMs differ only in their program

Differences to real computers:

- RAM has *infinitely many* registers
- RAM registers have *unlimited number of bits*
  (can store arbitrarily large natural numbers)
- more *technical details*

When creating the model, we abstract from the specific number of registers and bits per register.

Working mechanism of a RAM:

- works *in cycles*
- exactly one instruction is executed in each cycle (the one whose number is in the instruction register IR)
- Start: 0 is in IR, i.e., we start at the instruction with number 0
- Stop: if the instruction register points to the stop instruction or there is no instruction with the corresponding number

$[\mathrm{R}i]$ = Contents of register $\mathrm{R}i$
$[\mathrm{IR}]$ = Contents of register IR

Definition 2.28 (modified difference)

The function $\mathrm{md}\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ is defined as follows.

$$x \dot{-} y = \mathrm{md}(x, y) \stackrel{df}{=} \begin{cases} x - y, & \text{if } x > y \\ 0 & \text{otherwise.} \end{cases}$$

The following instructions can be used in RAM programs:

Instructions (syntax)	**Effect** (semantics)
Transport instructions	
$\mathsf{R}i \leftarrow \mathsf{R}j$ (direct address)	$[\mathsf{R}i] := [\mathsf{R}j]$, $[\mathsf{IR}] = [\mathsf{IR}] + 1$
$\mathsf{R}i \leftarrow \mathsf{RR}j$ (indir. Address.)	$[\mathsf{R}i] := [\mathsf{R}[\mathsf{R}j]]$, $[\mathsf{IR}] = [\mathsf{IR}] + 1$
$\mathsf{RR}i \leftarrow \mathsf{R}j$ (indir. address.)	$[\mathsf{R}[\mathsf{R}i]] := [\mathsf{R}j]$, $[\mathsf{IR}] := [\mathsf{IR}] + 1$
Arithmetic instructions	
$\mathsf{R}i \leftarrow k$	$[\mathsf{R}i] := k$, $[\mathsf{IR}] = [\mathsf{IR}] + 1$
$\mathsf{R}i \leftarrow \mathsf{R}j + \mathsf{R}k$	$[\mathsf{R}i] := [\mathsf{R}j] + [\mathsf{R}k]$, $[\mathsf{IR}] = [\mathsf{IR}] + 1$
$\mathsf{R}i \leftarrow \mathsf{R}j - \mathsf{R}k$	$[\mathsf{R}i] := [\mathsf{R}j] \dot{-} [\mathsf{R}k]$, $[\mathsf{IR}] = [\mathsf{IR}] + 1$

Instructions (syntax)	**Effect** (semantics)
Jump instructions	
GOTO m	$[\mathrm{IR}] := m$
IF R$i = 0$ GOTO m	$[\mathrm{IR}] := \begin{cases} m, & \text{if } [\mathrm{R}i] = 0 \\ [\mathrm{IR}] + 1 & \text{else} \end{cases}$
IF R$i > 0$ GOTO m	$[\mathrm{IR}] := \begin{cases} m, & \text{if } [\mathrm{R}i] > 0 \\ [\mathrm{IR}] + 1 & \text{else} \end{cases}$
Stop instruction	
STOP	end of computation

Example 2.29

Computation of the multiplication using a RAM: $x \cdot y$

Starting situation:	$[\mathrm{R0}] = x$, $[\mathrm{R1}] = y$, $[\mathrm{R}i] = 0$ for $i \geq 2$
Final situation:	$[\mathrm{R0}] = x \cdot y$
Auxiliary register:	$[\mathrm{R2}] =$ intermediate total, $[\mathrm{R3}] = 1$
Idea:	$x \cdot y = \underbrace{x + x + \cdots + x}_{y\text{-times}}$

Program:

0	$\mathrm{R3} \leftarrow 1$
1	IF $\mathrm{R1} = 0$ GOTO 5
2	$\mathrm{R2} \leftarrow \mathrm{R2} + \mathrm{R0}$
3	$\mathrm{R1} \leftarrow \mathrm{R1} - \mathrm{R3}$
4	GOTO 1
5	$\mathrm{R0} \leftarrow \mathrm{R2}$
6	STOP

For the computation $5 \cdot 3$ we denote the situation after cycle i:

Cycle i	[IR]	instruction to be executed	[R0]	[R1]	[R2]	[R3]
	0	$R3 \leftarrow 1$	5	3	0	0
1	1	IF $R1 = 0$ GOTO 5	5	3	0	1
2	2	$R2 \leftarrow R2 + R0$	5	3	0	1
3	3	$R1 \leftarrow R1 - R3$	5	3	5	1
4	4	GOTO 1	5	2	5	1
5	1	IF $R1 = 0$ GOTO 5	5	2	5	1
6	2	$R2 \leftarrow R2 + R0$	5	2	5	1
7	3	$R1 \leftarrow R1 - R3$	5	2	10	1
8	4	GOTO 1	5	1	10	1
9	1	IF $R1 = 0$ GOTO 5	5	1	10	1
10	2	$R2 \leftarrow R2 + R0$	5	1	10	1
11	3	$R1 \leftarrow R1 - R3$	5	1	15	1
12	4	GOTO 1	5	0	15	1
13	1	IF $R1 = 0$ GOTO 5	5	0	15	1
14	5	$R0 \leftarrow R2$	5	0	15	1
15	6	STOP	15	0	15	1

Example 2.30

Transfer maximum of $[R10], [R11], [R12], \ldots, [R[R1]]$ to R0.

Starting situation: input is in R1, R10, R11, ..., R[R1],
Content of other registers is arbitrary

Idea: Consider Ri for
$i = [R1], [R1] - 1, \ldots, 12, 11, 10$;
the maximum of [Ri] and [R0]
is moved to R0

```
0  R0 ← 0
1  IF R1 ≤ 9  GOTO 6        // disallowed instruction
2  IF RR1 ≤ R0  GOTO 4      // disallowed instruction
3  R0 ← RR1
4  R1 ← R1 − 1              // disallowed instruction
5  GOTO 1
6  STOP
```

We are now eliminating the disallowed instructions:

0	0	$R0 \leftarrow 0$	
1	1	$R2 \leftarrow 9$	
	2	$R3 \leftarrow R1 - R2$	$(a \leq b \Leftrightarrow a \dot{-} b = 0)$
	3	IF $R3 = 0$ GOTO 11	
2	4	$R3 \leftarrow RR1$	
	5	$R3 \leftarrow R3 - R0$	$(a \leq b \Leftrightarrow a \dot{-} b = 0)$
	6	IF $R3 = 0$ GOTO 8	
3	7	$R0 \leftarrow RR1$	
4	8	$R2 \leftarrow 1$	
	9	$R1 \leftarrow R1 - R2$	
5	10	GOTO 1	
6	11	STOP	

RAMs provide a second mathematical characterization of the term *computability*.

Definition 2.31 (RAM computability)

- Let M be a RAM. A function $\varphi\colon \mathbb{N}^n \to \mathbb{N}$ with $n \geq 0$ **is computed by M** if for all $x_1, \ldots, x_n \in \mathbb{N}$:

$$\varphi(x_1, \ldots, x_n) = \begin{cases} y, & \text{if } M \text{ starting with } [\mathrm{IR}] = 0,\ [\mathrm{R0}] = x_1,\ \ldots,\ [\mathrm{R}(n-1)] = x_n,\ [\mathrm{R}i] = 0 \text{ for } i \geq n \text{ stops after a finite number of steps with } [\mathrm{R0}] = y \\ \text{n.d.} & \text{else} \end{cases}$$

- A function $\varphi : \mathbb{N}^n \to \mathbb{N}$ with $n \geq 0$ is called **RAM-computable**, if there is a RAM that computes φ.
- **RAM** $\stackrel{df}{=} \{\varphi \mid \varphi : \mathbb{N}^n \to \mathbb{N} \text{ is RAM-computable}\}$.

Remark 2.32

Each RAM computes an infinite number of functions, namely for every $n \geq 0$ exactly one n-ary function. For example, the RAM from Example 2.29 computes among others

- *the 2-ary function* $\mathrm{prod} : \mathbb{N}^2 \to \mathbb{N}$ *with* $\mathrm{prod}(x, y) \stackrel{df}{=} x \cdot y$,
- *the 1-ary function* $C_0^1 : \mathbb{N} \to \mathbb{N}$ *with* $C_0^1(x) \stackrel{df}{=} 0$ *and*
- *the 3-ary function* $f : \mathbb{N}^3 \to \mathbb{N}$ *with* $f(x, y, z) \stackrel{df}{=} (x \cdot y) + z$.

Encoding of Lists of Integers in While Programs

We will later show RAM = WHILE. To do this, we will first consider how to represent and process lists *of integers* in While programs and process them.

With the help of $\text{code}_{\mathbb{Z}}$ and the list encoding of natural numbers a list of integers $(y_1, \ldots, y_n)$ can be encoded by the following natural number:

$$\langle y_1, \ldots, y_n \rangle \stackrel{df}{=} \langle \text{code}_{\mathbb{Z}}(y_1), \ldots, \text{code}_{\mathbb{Z}}(y_n) \rangle.$$

The en- and decoding of such lists can be implemented using corresponding While programs. Therefore, the computation power does not change if the use of lists is permitted in While programs.
We are now adding lists and other language constructs to While programs. These make programming easier but do not change the computation power. We call the extended programs *Python programs*.

Python Programs

- All constructs from While programs are allowed
- Allows the use of lists (see next page)
- No strict bracketing of expressions and conditions
- multiplication $*$, division //, modulo % are allowed (correspond to the functions prodZ, divZ, modZ)
- functions without a return value are permitted
- All constructs for which we have shown that they can be simulated in While programs are allowed

Theorem 2.33

A function $f : \mathbb{Z}^n \to \mathbb{Z}$ or $f : \mathbb{N}^n \to \mathbb{N}$ is While-computable if and only if it can be computed by a Python program.

Proof.

All new language constructs in Python programs can be simulated in While programs. □

Usage of Lists in Python Programs

- Lists are denoted in the form `[3,-7,8]`
- `[ ]` creates the empty list
- `a = [3,-7,8]` creates the list `[3,-7,8]` and assigns `a` the *reference* to this list
- `b = a` also assigns `b` the *reference* to `[3,-7,8]` (so `a` and `b` point to the same object)
- `a[2]` is the 2nd element of the list, where the count starts at 0 (e.g. `b = a[2]` or also `a[2]=10000`)
- `a[3:6]` creates the partial list from the 3rd to 5th element of `a` , first index is at 0 (e.g. `b=a[3:6]` or `a[3:6]=[3,4,5]`)
- `len(a)` returns the number of elements in the list `a`
- `a + b` generates the list that is created by writing the lists `a` and `b` one behind the other
- `a += b` appends the elements of the list `b` to the list `a`
- Details: Python Language Reference

Example 2.34 (Usage of Lists)

The **Fibonacci function** $\mathrm{fib} : \mathbb{Z} \to \mathbb{Z}$ is inductively defined by:

$$\mathrm{fib}(n) \stackrel{df}{=} \begin{cases} 1, & \text{if } n \leq 1 \\ \mathrm{fib}(n-1) + \mathrm{fib}(n-2) & \text{otherwise} \end{cases}$$

Named after the Italian mathematician Fibonacci (around 1170–1240), who was looking for a formula for the multiplication of rabbits. His model:

- Initially (after 0 months) there is one female rabbit.
- After 2 months, a female rabbit has babies for the first time and then again every month.
- There is exactly one female young in each litter.
- Female rabbits never die.

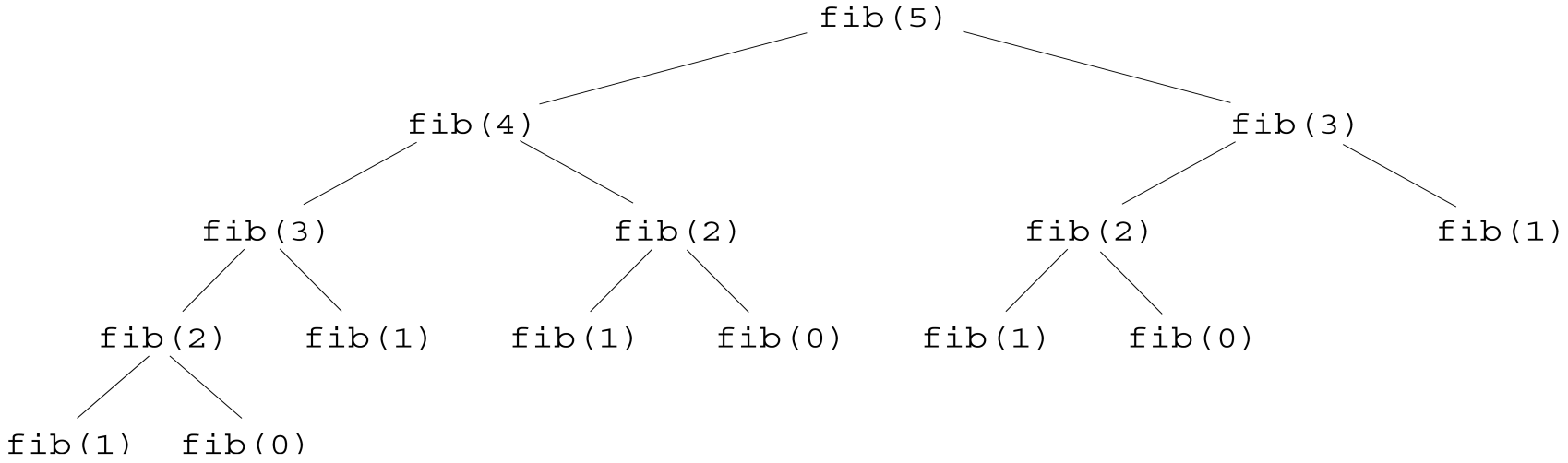
For $n \geq 0$ you can easily compute:

$$\text{fib}(n) = \text{number of female rabbits after } n \text{ months.}$$

The following Python program computes the Fibonacci function.

```
def fib(n):
  if (n <= 1):
    n = 1
  else:
    n = fib(n - 1) + fib(n - 2)
  return n
```

Two self-calls, which in turn cause self-calls again:



Short algorithm, but multiple calls (e.g. 5 times `fib(1)`).
The following Python program is much faster.

```
def fibFast(n):
  f = [1,1]                      # list [fib(0), fib(1)]
  if (n <= 1):                   # handle negative inputs
    n = 1
  for i in range(2,n+1):         # i = 2,3,...,n
    f += [f[i-1] + f[i-2]]       # append fib(i) to the list
  return f[n]                    # read fib(n) from the list
```

Non-recursive algorithm. Each value is only computed once:

$$\texttt{f[0]} \rightarrow \texttt{f[1]} \rightarrow \texttt{f[2]} \rightarrow \texttt{f[3]} \rightarrow \texttt{f[4]} \rightarrow \texttt{f[5]}$$

Storing and reusing intermediate solutions is also known as *dynamic programming*.

Theorem 2.35

RAM $\subseteq$ WHILE.

Proof idea:

- Step-by-step simulation of the RAM, whereby the registers are stored in two lists u and v

Proof.

Let $\varphi \in$ RAM and $\varphi : \mathbb{N}^n \to \mathbb{N}$.

Assume φ is computed by the following RAM M:

$$\begin{array}{cc} 0 & b_0 \\ 1 & b_1 \\ \vdots & \vdots \\ k & b_k \end{array}$$

W.l.o.g.: $b_k = \text{STOP}$ and $b_i \neq \text{STOP}$ for $i < k$.

We simulate M with a Python program:

- the variable `ir` stores contents of instruction register IR
- the list `u` contains the indices and the list `v` contains the corresponding contents of the previously used registers
- if a register Ri that has not yet been used is accessed, then i is appended to u and 0 to v (i.e. $[\mathrm{R}i] = 0$)

Example: $u = [5, 2]$ and $v = [24, 7]$ means that R5 and R2 have been used so far, $[\mathrm{R5}] = 24$ and $[\mathrm{R2}] = 7$.

We use the following functions "read" and "write" to access the Ri registers. They perform the search in the lists and extend the lists if necessary.

Attention: "read" and "write" change the lists visible in the main program, as lists are passed to functions by reference.

```
def read(u,v,a):                        # returns the content of Ra
  i = 0
  while (i < len(u) and u[i] != a): # search for index a
    i = i + 1
  if (i == len(u)):                     # Extend lists
    u += [a]
    v += [0]
  return v[i]                           # returns content of Ra

def write(u,v,a,b):                     # write b to Ra
  i = 0
  while (i < len(u) and u[i] != a): # search for index a
    i = i + 1
  if (i == len(u)):                     # Extend lists
    u += [a]
    v += [0]
  v[i] = b                              # write b in Ra
```

```
def phi(x1,x2,...,xn):
   u = [0,1,...,n-1]
   v = [x1,x2,...,xn]
   ir = 0
   while (ir < k):
        if (ir == 0):
            s_0
        if (ir == 1):
            s_1
                 ⋮
        if (ir == k - 1):
            s_{k-1}
    return read(u,v,0)
```

simulation of at least one cycle of M

The instruction block s_i simulates the RAM instruction b_i as follows:

If b_i is instruction ...	then s_i is instruction block ...

$\mathsf{R}a \leftarrow \mathsf{R}b$	`i = read(u,v,`b`)`
	`write(u,v,`a`,i)`
	`ir = ir + 1`

$\mathsf{R}a \leftarrow \mathsf{RR}b$	`i = read(u,v,`b`)`
	`j = read(u,v,i)`
	`write(u,v,`a`,j)`
	`ir = ir + 1`

$\mathsf{RR}a \leftarrow \mathsf{R}b$	`i = read(u,v,`b`)`
	`j = read(u,v,`a`)`
	`write(u,v,j,i)`
	`ir = ir + 1`

$\mathsf{R}a \leftarrow b$	`write(u,v,`a`,`b`)` `ir = ir + 1`

$\mathsf{R}a \leftarrow \mathsf{R}b + \mathsf{R}c$	`i = read(u,v,`b`) + read(u,v,`c`)`
	`write(u,v,`a`,i)`
	`ir = ir + 1`

If b_i is instruction ...	then s_i is instruction block ...

$\mathrm{R}a \leftarrow \mathrm{R}b - \mathrm{R}c$	`i = read(u,v,`b`) - read(u,v,`c`)`
	`if (i < 0):`
	`  i = 0`
	`write(u,v,`a`,i)`
	`ir = ir + 1`

GOTO a	`ir =` a

IF $\mathrm{R}a = 0$ GOTO b	`if (read(u,v,`a`) == 0):`
	`  ir =` b
	`else:`
	`  ir = ir + 1`

IF $\mathrm{R}a > 0$ GOTO b	`if (read(u,v,`a`) > 0):`
	`  ir =` b
	`else:`
	`  ir = ir + 1`

Thus $D_{\psi_{\mathtt{phi}}} \cap \mathbb{N}^n = D_\varphi$ and $\psi_{\mathtt{phi}}(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
Then $\varphi \in$ WHILE by Definition 2.17. $\square$

2.5
Mini-While Programs

Learning Objectives

After this section you should

1. know the main differences between While programs and Mini-While programs.
2. be able to outline the translation of While programs into RAMs.

Goal: we want to prove WHILE $\subseteq$ RAM.

Idea: we take an intermediate step via a variant of While programs with a reduced range of functions
While program $\Rightarrow$ Mini-While program $\Rightarrow$ RAM program

The generation of intermediate code is also carried out with real compilers in order to

- simplify the translation process,
- be able to optimize better, and
- increase the portability of the compiler.

Syntax of Mini-While-Programs

1. **Identifiers, constants, variables:** as for While-Programs
2. **Expressions:**
   (a) a is constant or variable $\Rightarrow$ a is expression
   (b) a, b are variables $\Rightarrow$ $(a + b)$, $(a - b)$ are expressions
3. **Conditions:** a is variable $\Rightarrow$ $(a > 0)$ is condition

Syntax of Mini-While-Programs ctd.

4. **Instruction blocks:** (inductive definition)

   BC *Assignments*

   - a is Variable, b is Constant $\Rightarrow$ a = b is instr. block
   - a, b, c are variables $\Rightarrow$ a = (b + c), a = (b - c) are instr. blocks

   IS *Compound instruction blocks*

   - $s_1, \ldots, s_n$ are instr. blocks $\Rightarrow$ $\left.\begin{array}{c} s_1 \\ \vdots \\ s_n \end{array}\right\}$ is instr. block
   - a is variable, s is instr. block $\Rightarrow$ $\left.\begin{array}{c} \texttt{while(a} > 0\texttt{)}: \\ s \end{array}\right\}$ is instr. block

5. **Function declarations:** same as for While programs.
6. **Mini-While programs:** consist of exactly one function decl. without initialization and without function calls.

Semantics of Mini-While Programs

same ;-)

Definition 2.36 (Mini-While Computability)

- $\varphi : \mathbb{Z}^n \to \mathbb{Z}$ with $n \geq 0$ is called **Mini-While computable**, if there is a Mini-While program P such that $D_{\psi_P} = D_\varphi$ and $\psi_P(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
- $\varphi : \mathbb{N}^n \to \mathbb{N}$ with $n \geq 0$ is called **Mini-While computable**, if there is a Mini-While program P such that $D_{\psi_P} \cap \mathbb{N}^n = D_\varphi$ and $\psi_P(x_1, \ldots, x_n) = \varphi(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_\varphi$.
- **MINIWHILE** $\stackrel{df}{=} \{\varphi \mid \varphi : \mathbb{N}^n \to \mathbb{N}$ is Mini-While computable$\}$.

Theorem 2.37

WHILE $\subseteq$ MINIWHILE.

Proof sketch.

We eliminate all constructs of While programs step by step in the subsequent order that are not allowed in Mini-While programs.

1. For-loops
2. If-instructions
3. Conditions not of the form $(a > 0)$, where a is a variable
4. Function calls and thus also the declaration of several functions in a program
5. Assignments that do not have the form $a = \mathtt{d}$, $a = (b + c)$ or $a = (b - c)$ for a constant d and variables b, c

The step 4 is the most difficult. Among others, recursive self-calls of functions must be eliminated. To do this, we store the variable assignments and return addresses of the nested computation in lists and simulate the calls of functions in a while-loop. The lists are encoded in a natural number using $\langle y_1, \ldots, y_n \rangle$. $\square$

Theorem 2.38

WHILE $\subseteq$ RAM.

Proof sketch.

- Let $g \in$ WHILE $\subseteq$ MINIWHILE with $g : \mathbb{N}^n \to \mathbb{N}$.
- There is a Mini-While program P with $D_{\psi_P} \cap \mathbb{N}^n = D_g$ and $\psi_P(x_1, \ldots, x_n) = g(x_1, \ldots, x_n)$ for all $(x_1, \ldots, x_n) \in D_g$.
- P can be simulated on a RAM as follows:
- Each variable is represented by 2 registers, which store the absolute value and the sign.
- Assignments of the Mini-While program can now be carried out directly on the registers.
- Signed addition/subtraction is simulated by corresponding RAM programs (case study for $x \geq 0$, $y \geq 0$, $|x| \geq |y|$).
- While-loops: simulated with conditional jump instructions.

$\square$

This means that the computations models considered so far are equivalent.

Corollary 2.39

RAM = WHILE = MINIWHILE.

Proof.

Follows from Theorems 2.35, 2.37, and 2.38. □

Corollary 2.40

Every RAM-computable function $\varphi : \mathbb{N}^n \to \mathbb{N}$ can be computed by a RAM without indirect addressing.

Proof.

Follows from Corollary 2.39 and the proof of theorem 2.38, since the simulation described there does not use indirect addressing. □